SauceDemo – Test Plan

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Status: Proposed baseline – team review pending

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Item	Details
Application under test	https://www.saucedemo.com
Project	DEPI Graduation Project – Automated Testing Framework for a Web Application (Sprint 1: Planning)
Coverage	Manual functional testing, Java UI automation (Selenium, POM, Cucumber BDD) and limited performance observation
Execution status	Planning only. No pass/fail results are claimed.

Purpose

Define a practical, traceable testing approach for the SauceDemo customer journey: login, product browsing, cart and checkout through order confirmation. The plan sets scope, case counts, ownership, entry/exit criteria and the sprint schedule. Assignments and dates are proposed and need team review before execution. Everything in this plan is limited to features the site actually provides.

1. Product analysis

SauceDemo is a demo e-commerce web app from Sauce Labs. The main flow is Login → Inventory → Product details → Cart → Checkout (information → overview → complete). The details below were noted from the public site and must be reconfirmed in the Sprint 2 exploratory session.

Account	Purpose in testing
standard_user	Main account for baseline behaviour
locked_out_user	Must be rejected at login
problem_user	Demo account with intentional UI/behaviour anomalies
performance_glitch_user	Demo account with intentionally slow responses
error_user	Demo account with intentional errors in cart/checkout actions
visual_user	Demo account with intentional visual differences

All accounts share one public demo password shown on the login page. It is not a confidential credential, but it is kept in a configuration file rather than inside test code.

The inventory has six products: Sauce Labs Backpack (29.99), Sauce Labs Bike Light (9.99), Sauce Labs Bolt T-Shirt (15.99), Sauce Labs Fleece Jacket (49.99), Sauce Labs Onesie (7.99) and Test.allTheThings() T-Shirt (Red) (15.99). Prices are in USD and are reconfirmed before execution.

2. Objectives, scope and counts

Objective	Measurable target
Functional coverage	42 unique UI cases on Chrome; four selected cases repeated on Firefox
Automation	Automate exactly 12 named UI cases from the same 42-case catalog; extra candidates need change control
Performance observation	4 records: response timing for standard_user and performance_glitch_user, plus one low-volume JMeter run
Traceability	All 16 requirements mapped to case IDs; an outcome recorded for every execution
Completion	Meet the criteria in section 9 within the proposed 20-working-day cycle

In scope	Key checks
Login / session	Valid login, invalid and empty credentials, locked-out user, logout, protected-page access
Inventory	Product list, four sort options, product details, back navigation
Cart	Add (list and details page), multiple items, remove, empty state, persistence while browsing
Checkout / order	Customer information validation, order overview, item total, tax and total, finish, cancel
Menu and demo accounts	All Items, About, Reset App State; behaviour of problem_user, error_user and visual_user
Performance	Response-time observation for login and inventory; one low-volume JMeter run

Out of scope: real payments; server code and database validation; mobile apps; admin features; security or penetration testing; stress or high-load testing of the public site; full accessibility and browser audits. Chrome is primary and Firefox is limited to four smoke checks. The site has no public API, registration, search or user profile, so none of these are planned. Footer social links open external sites and are excluded.

Counting rules

Measure	Count / interpretation
Unique UI cases	42: accounts/session 10 + inventory 8 + cart 7 + checkout/order 11 + menu/demo accounts 6
UI baseline executions	46: 42 Chrome + four Firefox repeats. Repeats are not new unique cases.
Performance records	4 (PERF-01 to PERF-04), counted separately from UI
Combined baseline	46 unique cases (42 UI + 4 PERF); 50 execution records (46 UI + 4 PERF)
Automation	12 of the 42 UI cases. Automated reruns are reported separately and not added to the baseline.

3. Approach, environment and test data

Activity	Execution approach
Exploration / design	Run an unstructured exploratory session on every page, confirm messages and prices, and record questions instead of inventing requirements. Use equivalence partitions for input checks.
Smoke	Before each cycle check login, inventory and cart availability. Suspend only affected testing if essentials fail.
Manual / regression	Run positive and negative cases; repeat stable cases after changes. Retest a defect only when a fix is available.
UI automation	Java JDK 17, Selenium WebDriver, TestNG and Maven. Page Object Model, explicit waits, independent setup, assertions and failure screenshots.
BDD	Cucumber feature files in Gherkin, mapped to the automation steps. The first feature file is written in Sprint 2.
Performance	Manual timing with browser DevTools plus one throttled JMeter run on the login page (see section 5).
Compatibility	Repeat UI-001, UI-011, UI-019 and UI-034 on Firefox. Record each browser result separately.
Defect tracking	Jira for test cases, defects and sprint progress.
Environment item	**Planned configuration**
Host / browsers	Team Windows laptops; Chrome primary; Firefox smoke. Record OS, browser and driver versions in reports.
Tools	Selenium, TestNG, Maven, Cucumber, JMeter, Jira, Git/GitHub, Jenkins (Sprint 4)
Collaboration	Git/GitHub with branches, pull requests and reviews; shared case workbook and defect log
Access	Public site only; no server or database credentials are assumed
Test data	The six demo accounts; fictitious checkout data (for example first name Test, last name Tester, postal code 11511); product names and prices from section 1
Isolation	Application state is held in the browser, so every case starts from a clean session or after Reset App State. No data is created on a server, so no cleanup of accounts is needed.

4. UI case catalog

Priorities: P1 = critical path, P2 = important, P3 = minor. Expected messages follow the wording noted from the public site and are confirmed during exploration. Each case is expanded into preconditions, data and steps in the case workbook before execution.

4.1 Accounts and session

ID	Req.	Scenario	Expected outcome	Pri.
UI-001	R01	Valid login (standard_user)	Inventory page opens with the product list	P1
UI-002	R01	Login with each other demo account except locked_out_user	Each account reaches the inventory page	P2
UI-003	R02	Wrong password	Login rejected: username and password do not match any user	P2
UI-004	R02	Unknown username	Same rejection message; user stays on the login page	P2
UI-005	R02	Empty username	Message: Username is required	P2
UI-006	R02	Empty password	Message: Password is required	P2
UI-007	R02	Both fields empty	Message: Username is required	P2
UI-008	R03	Login with locked_out_user	Rejected with a locked-out message	P1
UI-009	R04	Logout from the menu	Session ends and the login page is shown	P1
UI-010	R04	Open the inventory URL without logging in	Access denied with a message that login is required	P2

4.2 Inventory and product details

ID	Req.	Scenario	Expected outcome	Pri.
UI-011	R05	Inventory list	Six products, each with image, name, description, price and add button	P1
UI-012	R06	Sort Name (A to Z)	Products ordered alphabetically, ascending	P2
UI-013	R06	Sort Name (Z to A)	Products ordered alphabetically, descending	P2
UI-014	R06	Sort Price (low to high)	Products ordered by price, ascending	P2
UI-015	R06	Sort Price (high to low)	Products ordered by price, descending	P2
UI-016	R07	Open product details from the name	Details page of the selected product opens	P1
UI-017	R07	Details data matches inventory	Name, description, price and image match the list	P1
UI-018	R07	Back to products	Inventory page is shown again	P2

4.3 Cart

ID	Req.	Scenario	Expected outcome	Pri.
UI-019	R08	Add one product from inventory	Button changes to Remove; badge shows 1; item in cart with correct price	P1
UI-020	R08	Add several products	Badge equals number added; all items in cart with correct data	P1
UI-021	R08	Add from the details page	Item appears in cart and badge updates	P1
UI-022	R09	Remove from the inventory page	Button returns to Add; badge decreases	P2
UI-023	R09	Remove from the cart page	Selected item removed; other items unchanged	P1
UI-024	R10	Cart kept while navigating	Items remain after Continue Shopping and page changes	P2
UI-025	R10	Empty cart state	No items listed and no badge shown	P2

4.4 Checkout and order

ID	Req.	Scenario	Expected outcome	Pri.
UI-026	R11	Start checkout from cart	Customer information page opens	P1
UI-027	R11	Missing first name	Message: First Name is required	P2
UI-028	R11	Missing last name	Message: Last Name is required	P2
UI-029	R11	Missing postal code	Message: Postal Code is required	P2
UI-030	R11	Cancel on the information page	Returns to cart with items intact	P2
UI-031	R12	Overview content	Selected items, quantities, payment and shipping information shown	P1
UI-032	R12	Item total	Equals the sum of the item prices	P1
UI-033	R12	Tax and total	Total equals item total plus the displayed tax	P1
UI-034	R13	Finish the order	Confirmation message: Thank you for your order	P1
UI-035	R13	State after the order	Cart is empty; Back Home returns to inventory	P2
UI-036	R13	Cancel on the overview page	Returns to inventory; no order is placed	P2

4.5 Menu and demo accounts

ID	Req.	Scenario	Expected outcome	Pri.
UI-037	R14	Menu: All Items	Inventory opens from the cart or a details page	P2
UI-038	R14	Menu: Reset App State	Cart is emptied and the badge cleared	P2
UI-039	R14	Menu: About	The vendor website opens	P3
UI-040	R15	problem_user session (browse, add, checkout)	Observation case: behaviour matches standard_user; every deviation is documented	P2
UI-041	R15	error_user session (cart actions and checkout)	Observation case: same rule as UI-040	P2
UI-042	R15	visual_user session (layout and images)	Observation case: same rule as UI-040	P3

Firefox execution IDs: FX-01 = UI-001; FX-02 = UI-011; FX-03 = UI-019; FX-04 = UI-034. Same expectations as Chrome; four extra execution records, not new cases.

5. Performance observation cases

ID	Req.	Scenario	Method	Proposed target
PERF-01	R16	standard_user: login to inventory	Five runs timed in browser DevTools; use the average	Average ≤ 3 s
PERF-02	R16	performance_glitch_user: login to inventory	Same method; compare with PERF-01	Delay measured and documented against PERF-01
PERF-03	R16	standard_user: inventory page load after login	Five runs; use the average	Average ≤ 3 s
PERF-04	R16	JMeter low-volume run on the login page	5 virtual users, 60 seconds, simple GET requests	0% errors; average ≤ 3 s

The site documents no performance requirement, so the targets are proposals for team review. The JMeter run is deliberately small and limited to the login page; stress and load testing are not planned.

6. Workload and responsibilities

42 unique UI cases are divided equally: seven per member. The 12 automation cases are a subset of those cases, not extra coverage.

Member	Manual UI cases	Automation	Perf.	Supporting duties
Youssef Adel	UI-001 to UI-007: login and validation	UI-001, UI-003	–	Plan coordination, shared Java framework, GitHub integration
Ahmed Hossam El Din Ahmed	UI-008 to UI-014: locked user, logout, access control, list, three sorts	UI-009, UI-014	PERF-01	Jira setup for test cases and defects; review login scripts
Menna Mahmoud	UI-015 to UI-021: last sort, details, add to cart	UI-017, UI-020	–	Requirements matrix and Gherkin feature files
Ibrahim Mohamed	UI-022 to UI-028: remove, cart state, checkout start, information validation	UI-023, UI-026	PERF-02	Test data guidance; check cart arithmetic and defect-log completeness
Ahmed Ibrahim	UI-029 to UI-035: validation, overview, totals, order completion	UI-033, UI-034	PERF-04	JMeter script and Jenkins pipeline
Adham Amin	UI-036 to UI-042: cancel, menu, demo accounts	UI-036, UI-038	PERF-03	Documentation, final summary and Test Completion Report

Firefox: Youssef executes FX-01, Ahmed Hossam FX-02, Menna FX-03 and Ahmed Ibrahim FX-04. Every member writes and executes their assigned cases, records evidence and defects, and reviews a teammate.

7. Automation selection and ownership

Twelve UI cases are automated on Chrome. Each script prepares its own login and cart state and runs independently.

Case	Scenario	Owner	Reviewer
UI-001	Valid login	Youssef Adel	Ahmed Hossam El Din Ahmed
UI-003	Wrong password	Youssef Adel	Ahmed Hossam El Din Ahmed

Case	Scenario	Owner	Reviewer
UI-009	Logout	Ahmed Hossam El Din Ahmed	Menna Mahmoud
UI-014	Sort price low to high	Ahmed Hossam El Din Ahmed	Menna Mahmoud
UI-017	Details match inventory	Menna Mahmoud	Ibrahim Mohamed
UI-020	Add several products	Menna Mahmoud	Ibrahim Mohamed
UI-023	Remove from cart page	Ibrahim Mohamed	Ahmed Ibrahim
UI-026	Start checkout	Ibrahim Mohamed	Ahmed Ibrahim
UI-033	Tax and total	Ahmed Ibrahim	Adham Amin
UI-034	Finish the order	Ahmed Ibrahim	Adham Amin
UI-036	Cancel on overview	Adham Amin	Youssef Adel
UI-038	Reset App State	Adham Amin	Youssef Adel

Automation completion evidence

- A script with assertions exists for every listed case, built with the Page Object Model and mapped to Gherkin scenarios.
- Each case runs without relying on another test.
- A reviewed run report and failure screenshots are attached; a green build alone is not proof.
- Runs use separate IDs such as AUTO-UI-001 and are reported separately from the 50-record baseline.
- If a selected case becomes unsuitable, document the replacement, reason, owner and coverage impact before changing the baseline.

8. Requirements traceability matrix

Proposed requirement baseline derived from the visible behaviour of the site and the agreed scope. Mapping alone does not prove a pass.

Req.	Requirement	Linked cases
R01	Accept valid login for supported accounts	UI-001, UI-002; FX-01
R02	Reject invalid or incomplete credentials with a message	UI-003 to UI-007
R03	Block the locked-out account	UI-008
R04	End the session and protect inventory pages	UI-009, UI-010
R05	Display the product list	UI-011; FX-02
R06	Sort products by name and price	UI-012 to UI-015

Req.	Requirement	Linked cases
R07	Show product details consistent with the list	UI-016 to UI-018
R08	Add products to the cart	UI-019 to UI-021; FX-03
R09	Remove products from the cart	UI-022, UI-023
R10	Keep cart state and show the empty state	UI-024, UI-025
R11	Validate customer information at checkout	UI-026 to UI-030
R12	Show correct order overview and calculations	UI-031 to UI-033
R13	Complete or cancel the order	UI-034 to UI-036; FX-04
R14	Menu and navigation actions work	UI-037 to UI-039
R15	Document behaviour of the special demo accounts	UI-040 to UI-042
R16	Observe response time of login and inventory	PERF-01 to PERF-04

Before execution confirm that all 16 requirements have cases and all 46 unique cases have requirement links. After execution report requirements covered by design separately from requirements validated by passing results.

9. Execution control and completion criteria

Entry criteria: scope, 46 case definitions, 16 requirement mappings and expected outcomes reviewed; tools, test data and Jira project ready; site reachable; smoke checks pass; exploratory session completed and expected messages confirmed.

Suspension: suspend affected cases for a site outage, an unusable login, or a critical failure in a dependent flow. **Resumption:** resume after the site is available again and the relevant smoke check passes.

Exit criteria (proposed baseline)

- All 50 baseline records are executed: 42 Chrome + four Firefox + four performance. No Blocked or Not Run records remain.
- UI success is at least 44 of 46 browser-specific records (about 95%). All P1 cases pass on Chrome and all four Firefox checks pass.
- UI-040 to UI-042 are observation cases: they pass when the session is completed and every deviation is documented. Known intentional demo-account anomalies are logged as Known/Intentional, not as blocking defects.
- PERF-01 to PERF-04 are executed with measurements recorded; PERF-01, PERF-03 and PERF-04 meet the agreed targets.
- The 12 selected automated cases are implemented, reviewed and executed, and all 12 pass. A separate run report is published.
- No unresolved Critical or High defects; evidence, defects, traceability and final reports are reviewed.

Metric	Definition
Baseline completion	Executed / planned: UI 46/46; PERF 4/4. Executed means Pass or Fail, not Blocked or Not Run.
Baseline pass rate	UI passed / 46 × 100. Blocked records stay in the denominator.
Rerun treatment	Keep attempt history; use the latest reviewed outcome per planned execution ID. Retries are not new coverage.
Requirement coverage	Design coverage: 16/16 mapped. Execution and passing coverage come from linked case results.

Unmet criteria: report actual counts, incomplete coverage and open defects. Document each exception with affected IDs, severity, impact, rationale and team agreement, and label completion as conditional where applicable.

10. Defects and risks

Defect lifecycle (Jira): New → Reviewed → Open → Ready for retest → Closed or Reopened. Product defects, script failures and environment blocks are kept separate. Each report includes ID, title, environment, preconditions, steps, expected and actual results, severity, priority, evidence and the linked case.

Severity	Meaning
Critical	Core journey blocked with no workaround (login, add to cart or checkout unusable)
High	Major failure in login, cart, totals or order completion
Medium	Partial failure with a workaround (for example a wrong sort order)
Low	Cosmetic or minor text issue

Risk	Mitigation / owner
Site outage or slow network	Record blocks, resume after recovery; repeat timing runs. Youssef coordinates.
Browser-held state carries over between runs	Start each case from a clean session or Reset App State. All members.
Intentional anomalies of demo accounts mistaken for product defects	Log as Known/Intentional and review with the team. Adham and Ibrahim.
Unstable locators or timing in scripts	Review selectors, explicit waits and failure evidence. Framework and module owners.

Risk	Mitigation / owner
Limited time	Prioritise P1, record gaps and open issues honestly. Whole team.

11. Schedule and project roadmap

Proposed 20-working-day cycle aligned with the four DEPI sprints. Adjust to the official sprint calendar.

Sprint	Working days	Main activities	Deliverables
1 – Planning	Days 1–3	Confirm site, scope, assignments, entry/exit criteria, test schedule and roadmap	Test Plan, Project Roadmap
2 – Analysis and design	Days 4–8	Exploratory session; expand the 42 UI cases and 4 performance cases; log them in Jira; first Gherkin feature file; initial framework setup	Initial manual and automated test cases, Jira integration, first feature file
3 – Implementation	Days 9–13	GitHub repository and branching; POM scripts for the 12 cases; JMeter setup and low-volume script	Repository with documentation, automation scripts, JMeter script
4 – Test completion and execution	Days 14–20	Optimise the framework; execute the full suite; regression and retests; Jenkins pipeline; consolidated report; evaluate exit criteria	Optimised framework, execution report, final documentation, CI/CD pipeline, Test Completion Report

Agile practices (all sprints): daily stand-ups of at most 15 minutes (yesterday, today, blockers) and a retrospective at the end of each sprint (what went well, what did not, improvements).

12. Delivery checklist and change control

Updated plan; case and execution workbook; requirements matrix; Jira defect log with evidence; Java/Maven automation project with README; Cucumber feature files; JMeter script and results; Jenkins pipeline; execution report and Test Completion Report.

Scope, count or ownership changes are reviewed with all six members, and the schedule, dependencies and denominators are updated together.

References: https://www.saucedemo.com. Site details were noted from public use and must be rechecked before execution. This plan claims no executed results.